

EC 9540: Human Computer Interaction

LAB 02

Requirement Analysis and Hierarchical Task Analysis

Smart Hostel Maintenance Reporting System

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1. Introduction

This report presents the second stage of the Human Computer Interaction (HCI) design process for the Smart Hostel Maintenance Reporting System. Building on the target user identification and questionnaire prepared in Lab 1, a Google Forms questionnaire was distributed to undergraduate students residing in the university hostels. A total of 122 responses were collected. This report summarises the findings of that questionnaire, derives user requirements and expectations from the data, identifies the core task performed by the target users, and presents a Hierarchical Task Analysis (HTA) of the main goal, broken down into sub-tasks, actions, and a task execution plan.

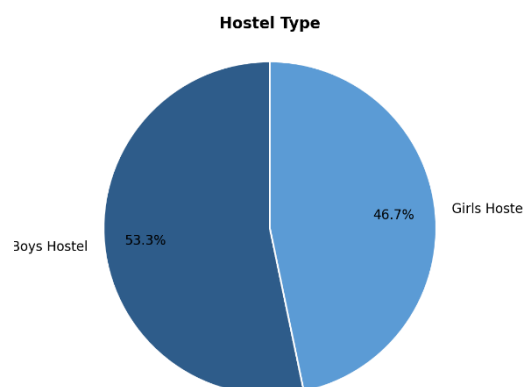
2. Target User Description

The target users of the Smart Hostel Maintenance Reporting System are undergraduate students residing in university hostels, drawn from three faculties: Engineering, Technology, and Agriculture. Separate hostels are maintained for male and female students. The system also serves the Warden, Sub Warden, Security personnel, and Maintenance Staff as secondary/administrative and operational users, as identified in Lab 1. For the purposes of this requirement analysis, the primary target user considered is the Hostel Student, since this is the user group who initiates the core task of the system - reporting a maintenance issue - and who was surveyed through the questionnaire.

2.1 Profile of Surveyed Respondents (n = 122)

Attribute	Distribution
Hostel	Boys' Hostel 53.3% Girls' Hostel 46.7%
Faculty	Engineering 43.4% Technology 37.7% Agriculture 18.9%
Year of Study	1st Year 41.8% 4th Year 40.2% 3rd Year 16.4% 2nd Year 1.6%
Duration in Hostel	6-12 Months 39.3% Less than 6 Months 31.1% More than 1 Year 29.5%
Currently residing in hostel	Yes 95.9% No 4.1%

The respondent pool is broadly representative of the hostel population across faculties, years of study, and length of residence, which supports the validity of the findings below.



3. Summary of Questionnaire Findings

3.1 Current (Manual) Reporting Process

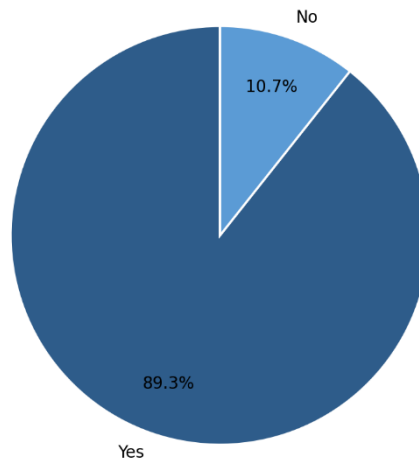
Finding	Result
Have ever reported a maintenance issue	89.3% Yes
Most common reporting channel	Hostel Office (42.6%), Warden (22.1%), WhatsApp (20.5%)
Frequency of maintenance issues	Very Frequently 33.6%, Weekly 31.1%, Monthly 18.9%, Rarely 16.4%
Most common issue types	Furniture Damage (34.4%), Light Issue (32.0%), Broken Fan (29.5%), Water Leak (28.7%)
Complaint eventually resolved	70.5% Yes / 29.5% No
Typical resolution time	2-3 Days 31.1%; More than a Week 20.5%; Never Resolved 18.9%
Average satisfaction with current process (1-5)	2.25 / 5
Average ease of reporting (1-5)	2.17 / 5
Average overall satisfaction (1-5)	2.22 / 5
Can track complaint status after reporting	Never 50.0%, Sometimes 34.4%, Always 15.6%
Avoided reporting an issue because process was difficult	75.4% Yes

Problems Reported with the Current Manual System

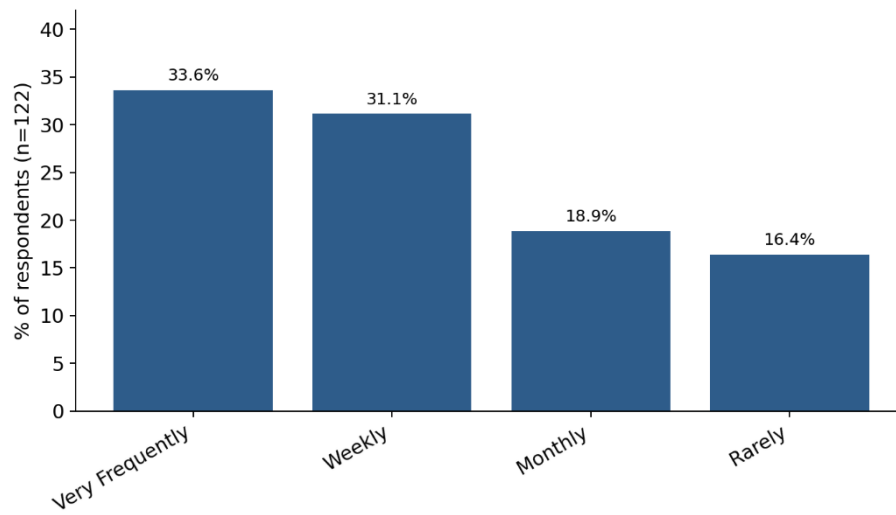
- No status updates on submitted complaints – 50.0%
- Difficult communication with staff – 49.2%
- No maintained repair history for a room/facility – 44.3%
- Complaints get lost – 42.6%
- Complaints get delayed – 40.2%
- Students must visit the hostel office in person – 36.9%

These findings closely confirm the problems identified qualitatively in Lab 1 (no digital record, lost complaints, no tracking, and repeated in-person visits), and quantify their prevalence among the student population.

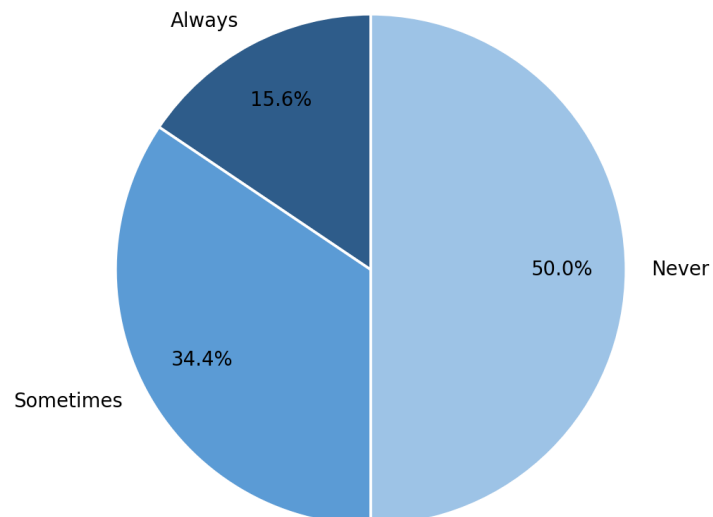
Ever Reported a Maintenance Issue



Frequency of Maintenance Issues

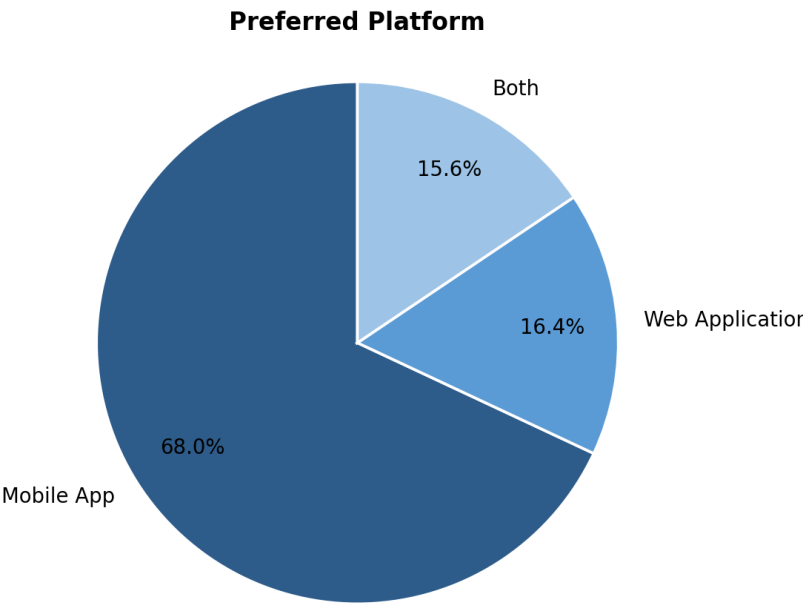


Ability to Track Complaint Status

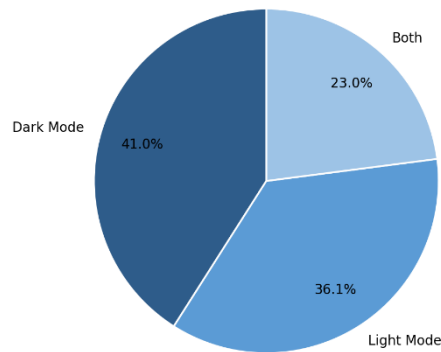


3.2 Expectations for the Proposed Digital System

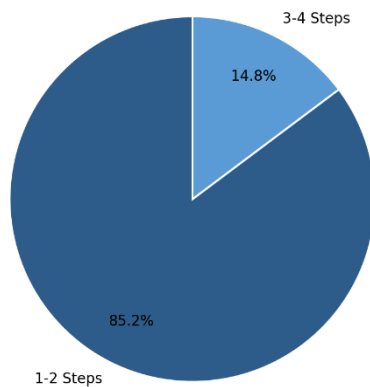
Finding	Result
Preferred platform	Mobile App 68.0%, Both 15.6%, Web Application 16.4%
Primary device used	Mobile Phone 82.0%, Laptop 13.9%, Tablet 2.5%, Desktop 1.6%
Preferred interface complexity	Very Simple 53.3%, Moderate 40.2%, Advanced 6.6%
Preferred colour theme	Dark Mode 41.0%, Light Mode 36.1%, Both (toggle) 23.0%
Importance of large buttons / clear icons (1-5)	Average 4.52 / 5 (63.9% rated 5)
Preferred language(s)	Sinhala 58.2%, Tamil 44.3%, English 41.8%
Preferred number of steps to submit a complaint	1-2 Steps 85.2%, 3-4 Steps 14.8%
Willingness to use the proposed system	Definitely Yes 74.6%, Probably Yes 22.1%, Not Sure 3.3%



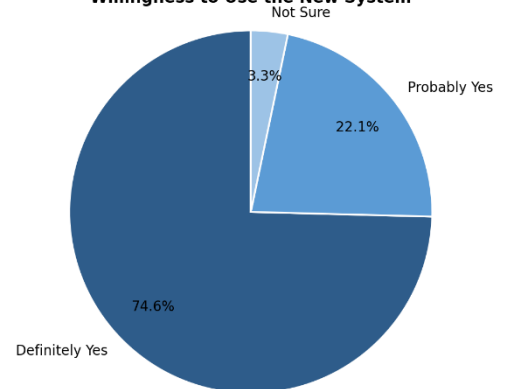
Preferred Colour Theme



Preferred Number of Steps to Submit



Willingness to Use the New System



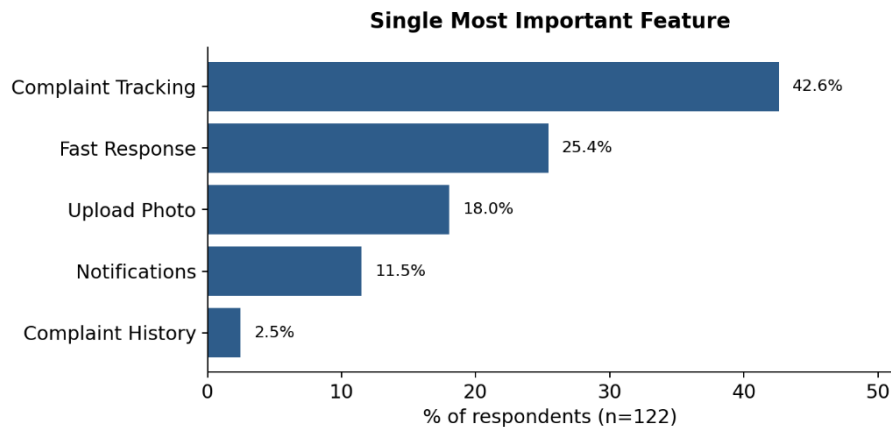
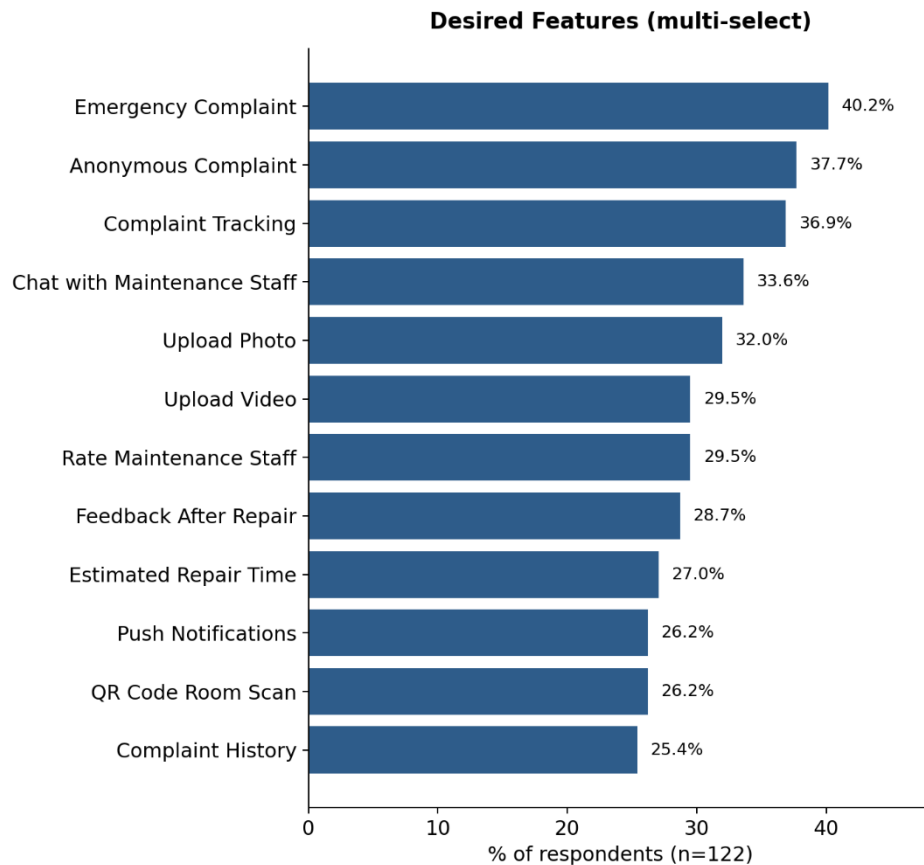
Most Requested Features (multi-select)

- Emergency Complaint flag – 40.2%
- Anonymous Complaint option – 37.7%
- Complaint Tracking – 36.9%
- In-app Chat with Maintenance Staff – 33.6%
- Upload Photo – 32.0%
- Upload Video, Rate Maintenance Staff – 29.5% each
- Feedback After Repair – 28.7%
- Estimated Repair Time / ETA, Push Notifications, QR Code Room Scan – 26–27% each

Single MOST Important Feature

- Complaint Tracking – 42.6%
- Fast Response – 25.4%
- Upload Photo – 18.0%
- Push Notifications – 11.5%
- Complaint History – 2.5%

Complaint tracking is, by a clear margin, the single feature students value most – consistent with the finding that 50% of students currently never know the status of their complaint (Section 3.1).



Common Suggestions (free-text responses)

- Implement real-time push notifications when complaint status changes.
- Provide an estimated repair completion time.
- Support dark mode for night-time use.
- Add a QR code on room doors to auto-fill room details.
- Provide an anonymous complaint option for sensitive issues.
- Prioritise faster response, especially during exam periods and for emergencies (e.g., major water leaks).
- Keep the submission form short (1-2 steps) and mobile-friendly.
- Support Sinhala and Tamil language options.

4. User Requirements and Expectations

4.1 Functional Requirements

1. The system shall allow a student to submit a maintenance complaint in no more than 1-2 steps, including room/location details, issue category, and optional photo/video evidence.
2. The system shall allow students to track the real-time status of a submitted complaint (e.g., Submitted, Assigned, In Progress, Resolved).
3. The system shall send push notifications automatically whenever a complaint's status changes.
4. The system shall allow a QR code fixed on each room door to auto-fill the room number when reporting an issue.
5. The system shall provide an option to flag a complaint as an emergency for priority handling.
6. The system shall provide an anonymous reporting option for sensitive complaints.
7. The system shall allow students to view an estimated repair time / ETA after a complaint is assigned.
8. The system shall allow students to rate maintenance staff and give feedback after a repair is completed.
9. The system shall maintain a complaint history per student and per room, so recurring issues can be identified.
10. The system shall provide an in-app chat channel between the student and the assigned maintenance staff.
11. The system shall be available as a mobile application (primary) with an optional web dashboard for administrative users.

4.2 Non-Functional (Usability/HCI) Requirements

12. The interface shall use a Very Simple layout by default (per the preference of 53.3% of respondents), with large, clearly labelled buttons and icons (rated 4.52/5 in importance).
13. The system shall support both Dark Mode and Light Mode, since preferences were split (41.0% / 36.1% / 23.0% wanting both).
14. The system shall support Sinhala, Tamil, and English, reflecting the multilingual composition of the student body (58.2% / 44.3% / 41.8% respectively).
15. The system shall be optimised primarily for mobile phones (82.0% of respondents' primary device), while remaining usable on laptops/desktops for administrative users.
16. The system shall minimise the number of steps and fields required to submit a complaint, to reduce the abandonment observed in the manual process (75.4% of students avoided reporting an issue at least once because the process was difficult).
17. The system shall provide clear, continuous visibility of complaint status to address the finding that 50.0% of students currently never know the status of a reported issue.

5. Core Task Performed by the Target User

Based on the questionnaire findings, the core task performed by the primary target user (the hostel student) is:

“Reporting a hostel maintenance issue and tracking it through to resolution.”

This is identified as the core task because: (1) 89.3% of respondents have already performed this task manually and are the direct source of the reported problems; (2) the single most valued feature is Complaint Tracking (42.6%), which is inseparable from the act of reporting; and (3) the two most significant pain points reported – no status updates (50.0%) and difficult communication (49.2%) – both arise directly from this task in its current, manual form.

6. Main Goal of the User

The main goal of the student user when interacting with the Smart Hostel Maintenance Reporting System is defined as:

Goal 0: Successfully report a hostel maintenance issue and track it through to resolution with minimal effort and full visibility of progress.

7. Breakdown of the Main Goal into Sub-Tasks

The main goal is decomposed into four sub-tasks (Plan 0: do 1-2-3 in sequence; 4 is performed once the complaint is resolved):

18. Access the reporting system (log in / open the app).
19. Submit a maintenance complaint.
20. Track the status of the submitted complaint until it is resolved.
21. Provide feedback once the repair is complete.

8. Actions Required to Complete Each Sub-Task

Sub-task 1: Access the reporting system

- Open the mobile app (or web application for administrative users).
- Log in using student credentials / hostel ID, or continue as an already-authenticated user.

Sub-task 2: Submit a maintenance complaint

- Tap “Report Issue” on the home screen.
- Scan the QR code on the room door to auto-fill the room/hostel/block, or select the location manually.
- Select the issue category (e.g., electrical, water leak, furniture, door lock).
- Attach a photo or short video of the issue (optional).
- Mark the complaint as an emergency, if applicable.
- Choose to submit anonymously, if preferred.
- Submit the complaint.

Sub-task 3: Track the status of the complaint

- Open the “My Complaints” section of the app.
- View the current status (Submitted / Assigned / In Progress / Resolved) and estimated repair time.
- Receive and read push notifications whenever the status changes.
- Use the in-app chat to contact the assigned maintenance staff if clarification is needed.

Sub-task 4: Provide feedback after repair

- Confirm that the reported issue has been resolved.
- Rate the maintenance staff and the quality/speed of the repair.
- Optionally add a comment describing the outcome.

9. Hierarchical Task Analysis (HTA)

The Hierarchical Task Analysis below formalises the decomposition in Sections 7 and 8 into a numbered goal/sub-goal/operation structure with execution plans, following standard HTA notation.

0. Report and track a hostel maintenance issue through to resolution

Plan 0: Do 1-2-3 in sequence, then do 4 once 3 confirms the issue is resolved.

1. Access the reporting system

Plan 1: Do 1.1, then 1.2.

- 1.1 Open the mobile app / web application
- 1.2 Log in with student credentials (or continue if already authenticated)

2. Submit a maintenance complaint

Plan 2: Do 2.1-2.2-2.3 in sequence. Do 2.4 and 2.5 if applicable/optional. Then do 2.6.

- 2.1 Select “Report Issue”
- 2.2 Identify the location (scan room QR code or select hostel/block/room manually)
- 2.3 Select the issue category/type
- 2.4 Attach a photo or video of the issue (optional)
- 2.5 Flag as emergency / choose anonymous submission (optional)
- 2.6 Submit the complaint

3. Track the complaint until resolved

Plan 3: Do 3.1, then repeat 3.2 and 3.3 until status = Resolved.

- 3.1 Open “My Complaints”
- 3.2 View current status, assigned staff, and estimated repair time
- 3.3 Receive push notification on status change; use in-app chat if clarification is needed

4. Provide post-repair feedback

Plan 4: Do 4.1, then 4.2.

- 4.1 Confirm the issue has been resolved
- 4.2 Rate the maintenance staff and submit feedback

This HTA will be used as the basis for the task modelling, interface design, and interaction flow in the next stage of the HCI design process.

10. Conclusion

This report analysed 122 questionnaire responses collected from hostel students to understand the current manual maintenance-reporting process and expectations for a digital replacement. The findings confirmed that the manual process suffers from poor tracking (50% of students never know their complaint status), difficult communication, and a high rate of complaints being lost or delayed – problems that led 75.4% of students to avoid reporting an issue at least once. Based on these findings, functional and non-functional requirements were derived, the core task (reporting and tracking a maintenance issue) and the main user goal were identified, and this goal was decomposed into sub-tasks, actions, and a formal Hierarchical Task Analysis. This analysis will guide the interface design and interaction flow of the Smart Hostel Maintenance Reporting System in the subsequent stages of the HCI design process.